

Revisiting time-varying dynamics in stock market forecasting: A multi-source sentiment analysis approach with large language model

Zhiqi Shao^a, Xusheng Yao^{b,*}, Feng Chen^c, Ze Wang^d, Junbin Gao^a

^a Business School The University of Sydney, NSW, Australia

^b School of Business Administration, Northeastern University, Shenyang, Liaoning, 110819, China

^c School of Mathematics and Statistics, University of New South Wales, NSW, Australia

^d Institute of Transport and Logistics, The University of Sydney, NSW, Australia

ARTICLE INFO

Keywords:

Sentiment analysis
Time series prediction
Seemingly unrelated regression
Dynamic linear models
Large language model
Decision system

ABSTRACT

This paper presents the Heterogeneous Dynamic Seemingly Unrelated Regression with Dynamic Linear Models (HD-SURDLM), an innovative framework for stock return prediction that combines cutting-edge sentiment analysis with dynamic financial modeling. The model integrates sentiment data from 2.5 million Twitter posts and various news sources, utilizing state-of-the-art sentiment analysis tools such as VADER, TextBlob, and RoBERTa. HD-SURDLM refines Gibbs sampling for enhanced numerical stability and efficiency while capturing cross-sectional dependencies across multiple assets such as a portfolio. The model consistently outperforms traditional methods like LSTM, Random Forest, and RNN in forecasting accuracy. Empirical results show a 1.02% improvement in 1-day horizon forecasts, a 0.42% gain for 20-day predictions, and a 0.36% increase for 50-day forecasts. By effectively merging public sentiment with dynamic asset modeling, HD-SURDLM offers substantial improvements in short- and long-term prediction accuracy. Its capacity to capture both cross-sectional insights and temporal dynamics makes it an invaluable tool for investors, traders, and financial institutions navigating sentiment-driven markets. HD-SURDLM not only enhances predictive accuracy but also provides a robust decision-support system for financial stakeholders.

1. Introduction

In the rapidly evolving landscape of financial markets, accurately predicting stock returns across multiple assets has become increasingly crucial for investors, traders, and financial institutions. The advent of big data and the proliferation of social media platforms have opened new avenues for market analysis, particularly through the lens of public sentiment [1]. However, harnessing this wealth of information for reliable stock market prediction — especially across a range of assets — remains a significant challenge in both academic research and practical applications.

The financial world is characterized by its complexity, volatility, and interconnectedness. Traditional approaches to stock market prediction often fall short in capturing these nuanced dynamics, particularly when attempting to forecast returns for multiple assets simultaneously. Many current models, including deep learning approaches, struggle to provide the necessary cross-sectional insights into multiple stocks while maintaining interpretability—a crucial factor for stakeholders seeking to understand the rationale behind predictions.

Existing models for stock return prediction, such as Long Short-Term Memory (LSTM) networks [2], Recurrent Neural Networks [RNN;3], Lasso regression [4], Multi-Layer Perceptron [MLP;5], Support Vector Regression [SVR;6], and Random Forest models [7], have been widely used. Recent studies have explored innovative combinations of these methods with sentiment analysis to enhance prediction accuracy, such as the combination of Lasso regression with sentiment analysis for stock price prediction [8]. LSTM models integrated with sentiment scores have shown promise in stock price forecasting [9–11]. RNN-based models incorporating sentiment data have demonstrated improved prediction capabilities [12,13]. SVR, when combined with sentiment analysis, has proven effective for stock price prediction [14]. MLP has been used both for direct stock price prediction [15] and as a tool for sentiment analysis in financial contexts [16]. Random Forest models, enhanced with sentiment analysis, have been developed for financial forecasting [17,18].

However, current approaches to stock market prediction, while diverse, face several limitations. Deep learning models, such as LSTM and

* Corresponding author.

E-mail addresses: zhiqi.shao@sydney.edu.au (Z. Shao), xshyao@outlook.com (X. Yao), feng.chen@unsw.edu.au (F. Chen), ze.wang@sydney.edu.au (Z. Wang), junbin.gao@sydney.edu.au (J. Gao).

<https://doi.org/10.1016/j.dss.2024.114362>

Received 16 April 2024; Received in revised form 28 October 2024; Accepted 31 October 2024

Available online 8 November 2024

0167-9236/© 2024 Elsevier B.V. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Table 1
Comparison of model features.

Features	HD-SURDLM	LSTM	RNN	Lasso	MLP	Random Forest	SVR	SURDLM
Interpretability	✓	×	×	×	×	✓	✓	✓
Dynamic adaptation (time-varying coefficients)	✓	✓	✓	×	×	×	×	✓
Cross-sectional insights	✓	×	×	×	×	×	×	×
Temporal dynamics	✓	✓	✓	×	×	×	×	✓
Uncertainty quantification (stabilized gibbs sampling)	✓	×	×	×	×	×	×	×

RNN, have shown considerable potential in capturing complex patterns within market data but often lack interpretability, making it difficult for stakeholders to understand the underlying rationale behind their predictions. On the other hand, traditional models like Lasso regression and SVR tend to struggle with dynamic adaptation, often failing to reflect the rapidly changing nature of financial markets. Furthermore, many of these models focus on predicting individual stock returns in isolation, limiting their ability to capture cross-sectional insights and interdependencies among multiple assets. Additionally, most of these models are not equipped to handle the temporal dynamics inherent in time-series financial data.

To tackle the challenge, the primary objectives of this research are (1) to explore the intricate relationship between sentiments expressed on social and news media platforms and stock returns and (2) to develop a robust model capable of capturing these complex, time-varying relationships in financial markets:

To achieve these goals, we propose a novel approach: the Heterogeneous Dynamic Seemingly Unrelated Regression with Dynamic Linear Models (HD-SURDLM). This model builds upon the traditional SURDLM framework [19] by incorporating a broader spectrum of sentiment data and enhancing model accuracy through an updated Gibbs sampling procedure. Our approach leverages the most popular sentiment analysis techniques, including VADER [20], Bag of Words (BOW), TextBlob, and RoBERTa [21], to capture the multifaceted nature of public opinion from social media and news sources.

The HD-SURDLM offers several key advantages over existing approaches (see Table 1):

- **Interpretability:** HD-SURDLM provides greater transparency compared to neural networks like LSTM [2] and RNN [3], offering clearer insights into the factors driving stock price predictions, which is crucial for financial forecasting.
- **Dynamic Adaptation:** Unlike Lasso regression [4], which assumes static relationships, HD-SURDLM uses time-varying coefficients, making it better suited for capturing the dynamic nature of financial markets.
- **Cross-Sectional Insights:** HD-SURDLM models multiple stocks simultaneously, capturing correlations and spillover effects across assets, providing a more comprehensive market view than single-stock models like MLP [5].
- **Temporal Dynamics:** While models like Random Forest [7] struggle with time-varying data, HD-SURDLM excels by accounting for the temporal evolution of stock prices, crucial for analyzing rapidly changing sentiment.
- **Uncertainty Quantification:** HD-SURDLM employs a full Bayesian approach, using enhanced Gibbs sampling for more efficient handling of high-dimensional data, providing robust uncertainty quantification, unlike vanilla SURDLM [19].

This novel approach represents a significant advancement in the field of financial forecasting, offering a robust framework that combines the strengths of statistical modeling with the power of sentiment analysis. Our study makes several key innovations:

- **Enhanced Gibbs Sampling Method:** We refine the Gibbs sampling procedure to improve parameter estimation accuracy, addressing numerical stability issues and simultaneously processing the multiple assets to capture the potential relationship, significantly boosting predictive performance as discussed in Section 4.2.
- **Novel Tweet Weighting System:** We introduce a sophisticated weighting mechanism for tweets based on user influence and tweet attributes, enhancing the relevance and impact of social media data in our predictions, detailed in Section 2.
- **Integration of Multiple Sentiment Analysis Techniques:** We employ a comprehensive suite of sentiment analysis methods, including VADER, Bag of Words, TextBlob, and RoBERTa, to capture the nuanced sentiments expressed in social media and news.
- **Extensive Data Utilization:** Our model leverages an unprecedented dataset comprising over 2.5 million Twitter posts and Reuters news headlines, spanning from 2014 to 2021, providing a rich tapestry of public sentiment and market information

The remainder of this paper is structured as follows: The description of data in Section 2. Section 3 covers data engineering, while the methodology is detailed in Section 4. The experiment is presented in Section 5, with discussion and implementation in Section 6. Finally, the conclusion is provided in Section 7.

2. Data collection and description data

We collected daily trading data from Yahoo Finance, including prices and volumes, and derived features such as the standard deviation of adjusted closing prices, daily returns, and average returns over the previous three and seven days. Missing data points were imputed using values from prior trading days, though this approach is not ideal. Reuters News headlines were scraped using Python, with special attention to companies like Tesla and Facebook, where CEO mentions were included for sentiment analysis. Aggregated sentiment scores were calculated daily, as shown in Fig. 1. For Twitter data, we used the API to collect tweets containing stock tickers, constrained by retrieval limits, and computed daily sentiment scores as time-varying variables $X_{j,t}$, covering tweets from June 2014 to May 2021 for 20 stocks.

3. Data engineering

3.1. Twitter sentiment construction and sentiment scores

In this section, we outline our approach to Twitter account classification through attribute analysis in Section 3.1. Additionally, we describe the methodology for selecting sentiment scores in Section 3.1, and furnish details about the dataset in Section 2.

Feature attribute classification in Twitter data. Before we start the choosing of sentiment scores, we estimate the 99% quantile for each attribute to be the following (see Table 2) for the sample of tweets on S&P 500 from the year 2018 to 2020.

We define a tweet as influential if any of its metrics — likes, quotes, replies, or retweets — exceed the 99th percentile for those metrics. A tweet is considered to originate from an influential user

Data Engineering on Reuter News Data Engineering on Yahoo Finance

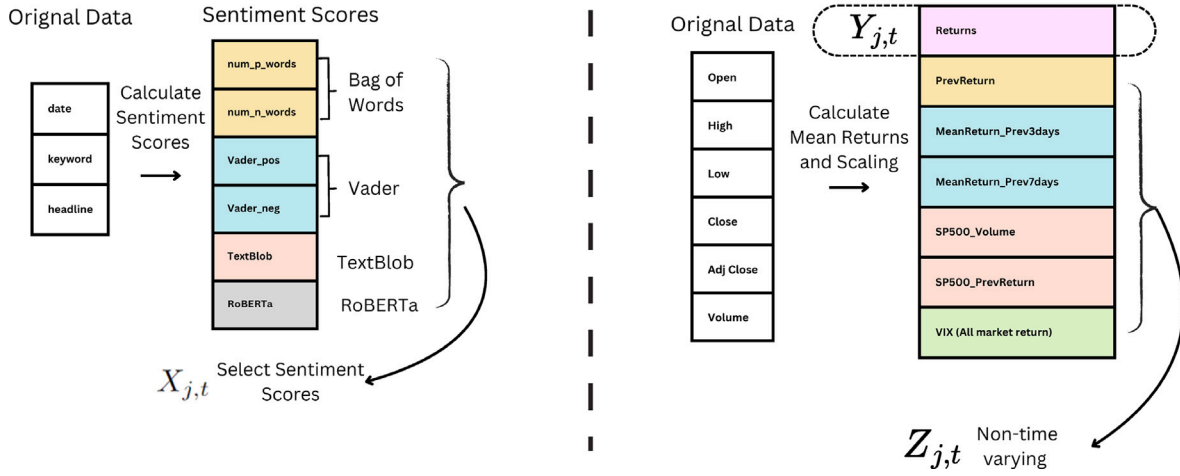


Fig. 1. The left figure depicts the data processing workflow for Reuter News. Initially, sentiment scores are calculated from the original data, subsequently enabling the selection of appropriate sentiments, denoted as $X_{j,t}$. Conversely, the right figure elucidates the data processing approach for Yahoo Finance. Initially, the mean return is computed and scaled. Subsequently, “Return” is utilized as a response variable, while the remaining variables, represented by $Z_{j,t}$, embody finance data features.

Table 2

Estimated 99% quantiles (rounded) of tweet attributes for S&P500.

Tweet attribute	99% quantile
Likes	100
Quotes	10
Replies	10
Retweets	10
Number of followers	1000

if the user has more than 1000 followers. For each sentiment score, we calculate average values across four distinct categories: influential tweets, non-influential tweets, tweets from influential users, and tweets from non-influential users. It is important to note that the categories of influential and non-influential tweets are comprehensive and mutually exclusive, yet they do not correlate directly with the categories of influential and non-influential users, which are also mutually exclusive and comprehensive.

Choice of sentiment scores. On top of the Tweeter account classification, we further use three methods to compute sentiment scores: Bag of Words (BoW), VADER, TextBlob (TB), and RoBERTa. TB provides only one score in the range of -1 to $+1$, so it can be used readily as a sentiment score, while both BoW and VADER yield two scores. For VADER, we subtract the negative sentiment scores (positive values) from positive sentiment scores for each provided corpus. The computation of BoW is:

$$\text{BoW score} = \begin{cases} \frac{\text{count}_p - \text{count}_n}{\text{count}_p + \text{count}_n}, & \text{if } \text{count}_p + \text{count}_n > 0; \\ 0, & \text{otherwise;} \end{cases} \quad (1)$$

where count_p and count_n denote the count of positive and negative words, respectively. Large Language Models (LLMs) have been widely adopted for sentiment analysis across various fields, including applications for time series data [22]. For LLM financial sentiment analysis tools, we compared FinBERT [23] and RoBERTa [21]. While FinBERT is well-suited for structured financial documents such as bank statements and formal reports, it may not perform effectively with noisy, large-scale datasets like Twitter. Given our need to handle more diverse and informal data, we selected RoBERTa as the preferred tool for this task due to its robustness in handling such data. In RoBERTa, we leverage a pre-trained RoBERTa model fine-tuned for sentiment analysis.

RoBERTa is a transformer-based model that provides sentiment scores by evaluating the context of the entire input text rather than relying solely on individual word counts or lexicons. The sentiment score is derived by passing the tweet text through the RoBERTa model, which outputs a probability distribution over sentiment classes (positive, neutral, and negative). The final sentiment score is computed by mapping these probabilities onto a continuous scale, similar to how VADER outputs a compound score. Specifically, it is computed as follows:

$$\text{RoBERTa sentiment score} = P(\text{positive}) - P(\text{negative})$$

where $P(\text{positive})$ and $P(\text{negative})$ represent the probabilities assigned by the model to the positive and negative sentiment classes, respectively.

We aggregate individual tweet or news headline sentiment scores into daily sentiments. For news, daily averages are computed. For Twitter, we compute daily weighted averages, with the weight based on the user’s follower count. This accounts for varying tweet influence. The general weighted average sentiment for each type z is given by:

$$\text{General Weighted Avg}_z = \frac{\sum_{i=1}^T (F_i + 1) \times S_{i,z}}{\sum_{i=1}^T (F_i + 1)}, \quad (2)$$

$$\text{Tweet Influential}_z = \frac{\sum_{i=1}^T D_i \times S_{i,z}}{\sum_{i=1}^T D_i}, \quad (3)$$

$$\text{Tweet Non-Influential}_z = \frac{\sum_{i=1}^T (1 - D_i) \times S_{i,z}}{\sum_{i=1}^T (1 - D_i)}, \quad (4)$$

$$\text{Account Influential}_z = \frac{\sum_{i=1}^T A_i \times S_{i,z}}{\sum_{i=1}^T A_i}, \quad (5)$$

$$\text{Account Non-Influential}_z = \frac{\sum_{i=1}^T (1 - A_i) \times S_{i,z}}{\sum_{i=1}^T (1 - A_i)}, \quad (6)$$

where F_i denotes the followers count of the i th tweet’s user, $S_{i,z}$ denotes the sentiment score of the i th tweet for the z th sentiment type (BoW, Vader, TB, or RoBERTa), D_i denote the binary flag for tweet influence (either 0 or 1) for the i th tweet, and A_i denote the binary flag for account influence (either 0 or 1) for the i th tweet’s user. The data process for Twitter can be found in Fig. 2.

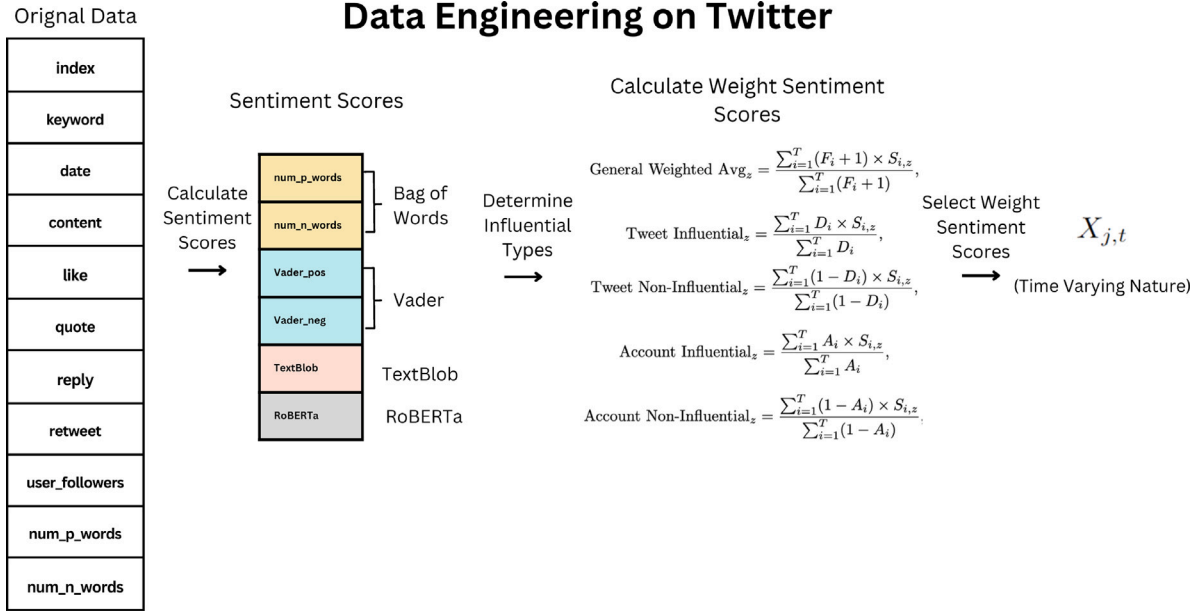


Fig. 2. The Twitter data will initially undergo sentiment analysis utilizing three distinct methodologies: Bag of Words, Vader, TextBlob and RoBERTa. Subsequently, influential types will be identified by applying Eqs. (2) to (6), and select the essential weight as time-varying nature variable for $X_{j,t}$. In the HD-SURDLM model, we select the general weight average as $X_{j,t}$.

3.2. Financial data processing

In our study, the focal point is the rate of return on stocks, denoted as y_{jt} for the j th stock at time t . In scenarios involving probability theory, Y may represent stock returns because of its non-stationary property. The rate of return at time t is calculated as $\frac{\text{Price}_{t+1} - \text{Price}_t}{\text{Price}_t}$, aiming to predict future prices based on present data, specifically using adjusted close prices to avoid forward-looking bias. Consequently, all predictive features are from the current or past times, ensuring predictions are for returns at least a day ahead. Features are categorized into non-time-varying, $Z_{j,t} \in \mathbb{R}^{p \times 1}$, and time-varying, $X_{j,t} \in \mathbb{R}^{q \times 1}$, with the latter including variables like sentiment scores. The process is shown in Fig. 1. This distinction is crucial for the HD-SURDLM hybrid model, allowing for the dynamic modeling of time-varying feature coefficients.

4. Method

In this section, we introduce the HD-SURDLM model in Section 4.1 and the stable Gibbs sampling in Section 4.2. The framework is showing as in Fig. 3.

4.1. HD-SURDLM model

To improve model performance, particularly given the use of multiple data sources and the incorporation of a stabilized sampling method, we propose a high-dimensional framework to incorporate multi-dimensional time-varying features. The extended model is articulated as:

$$y_{j,t} = \alpha_j + \left(\sum_{k=1}^K X_{j,t,k} \beta_{j,t,k} \right) + Z_{j,t} \gamma_j + u_{j,t}, \quad (7)$$

$$\beta_{j,t,k} = \beta_{j,t-1,k} + v_{j,t,k}, \quad (8)$$

where $j = 1, \dots, M$ denotes individual stocks; $t = 1, \dots, T$ represents discrete time points; and $k = 1, \dots, K$ indexes over the time-varying features. It comprises stock-specific intercepts α_j , time-varying features $X_{j,t,k}$ with coefficients $\beta_{j,t,k}$, and non-time-varying features $Z_{j,t}$ with coefficients γ_j . The model accounts for observation errors $u_{j,t} \sim \mathcal{N}_M(0, \Omega)$ and system errors $v_{j,t,k} \sim \mathcal{N}(0, \Omega)$, which are independent

across time and features, with a covariance structure $\Sigma = A\Omega A$. The unknown parameters $\Theta = \{\alpha_j, \beta_{j,t,k}, \gamma_j, \Omega\}_{j=1, \dots, M; t=1, \dots, T}$ collectively capture stock-specific effects, time-varying relationships, and error structures, enhancing the model's adaptability to complex, dynamic stock return patterns across multiple assets. The model utilizes collected data $D = \{y_{j,t}, X_{j,t,k}, Z_{j,t}\}$, enabling a comprehensive analysis of multi-asset stock returns while accounting for both time-varying and static influences (see Fig. 3).

4.1.1. Random parameters

In the specified HD-SURDLM model, delineated by Eqs. (7) and (8), several parameters are designated as random to encapsulate the inherent variability and dynamism in the stock returns. The parameter α_j serves as the intercept for each stock j , epitomizing the intrinsic level of returns in the absence of influence by other variables. The coefficients $\beta_{j,t,k}$ are pivotal, representing the time-varying impact of the k th predictor $X_{j,t,k}$ on the returns of stock j at time t . These coefficients may evolve as per Eq. (8) and are instrumental in portraying the temporal variations in the relationships between predictors and returns.

Additionally, the model incorporates γ_j , the set of coefficients corresponding to the predictor vector $Z_{j,t}$ for each stock j . Although these coefficients are static across time, their random nature allows for the modulation of the predictors' impact on returns. The parameter Ω , encompassed within the unknown parameter vector Θ , is not explicitly defined in the provided context but is potentially indicative of the variance-covariance matrix of the error terms, thereby capturing the variances and correlations in the model residuals.

Furthermore, the model accommodates error terms $u_{j,t}$ and $v_{j,t,k}$, representing the observation error and system error, respectively. The former accounts for the unexplained variability in stock returns, while the latter signifies the fluctuations in the time-varying regression coefficients, thereby contributing to the dynamic nature of the model. Collectively, the incorporation of these random parameters enhances the model's adaptability and proficiency in capturing the multifaceted relationships between various predictors and stock returns across diverse stocks and temporal instances.

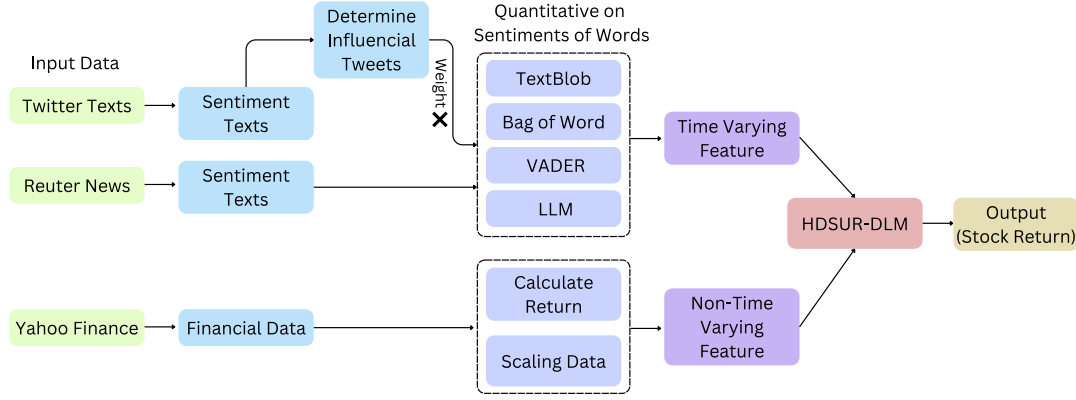


Fig. 3. This flowchart illustrates a comprehensive data analysis pipeline integrating sentiment data from Twitter and Reuters news with financial data from Yahoo Finance. The process begins with the extraction of sentiment from textual data, followed by the identification of influential tweets as a “weight”. It then quantitatively analyzes sentiment using methods like TextBlob, Bag of Words, VADER (tweet sentiment scores adjusted by their respective weights), and RoBERTa (LLM). Concurrently, financial returns are calculated and then normalized to the financial data. These steps create time-varying and non-time-varying features, which are then input into an advanced model, possibly denoted as HD-SURDLM. The model processes these inputs to generate the final output, which could inform financial decisions or forecasts.

4.2. Stabilized Gibbs sampling

We introduce a novel, stable Gibbs sampling approach that not only enhances performance but also effectively handles multiple assets, uncovering potential inter-asset relationships. Our method builds upon the work of Ho et al. [19], with key innovations designed to improve stability, efficiency, and multi-asset capability:

Sampling α and γ . We sample the intercept vector $\alpha = (\alpha_1, \dots, \alpha_M)$ and the non-time-varying coefficient matrix $\gamma = (\gamma_1, \dots, \gamma_M)$:

$$(\alpha|D, \Theta \setminus \{\alpha\}) \sim \mathcal{N}_M(\mu_\alpha, \Sigma_\alpha), \quad (9)$$

$$(\gamma|D, \Theta \setminus \{\gamma\}) \sim \mathcal{N}_{M \times p}(\mu_\gamma, \Sigma_\gamma), \quad (10)$$

where D represents the data, Θ is the set of all parameters, and p is the number of non-time-varying predictors. μ_α , Σ_α , μ_γ , and Σ_γ are derived from the full conditional distributions.

Sampling β . We sample the time-varying coefficient tensors $\beta = (\beta_{1,t,k}, \dots, \beta_{M,t,k})$ for each time-varying feature $k = 1, \dots, K$:

$$(\beta_{:, :, k}|D, \Theta \setminus \{\beta_{:, :, k}\}) \sim \mathcal{N}_{M \times T}(\mathbf{H}_k, \mathbf{B}_k). \quad (11)$$

We use an improved Filter Forward Backward Sampling (FFBS) algorithm that processes all M stocks concurrently:

1. Forward filtration: Compute $\mathbf{m}_t$ and $\mathbf{C}_t$ sequentially for $t = 1, \dots, T$.
2. Backward sampling: For $t = T, \dots, 1$, sample

$$\beta_t \sim \mathcal{N}_M(\mathbf{h}_t, \mathbf{B}_t), \quad (12)$$

where:

$$\mathbf{h}_t = \mathbf{m}_t + \mathbf{C}_t \mathbf{R}_{t+1}^{-1} (\mathbf{m}_{t+1} - \mathbf{m}_t), \quad (13)$$

$$\mathbf{B}_t = \mathbf{C}_t + \mathbf{C}_t \mathbf{R}_{t+1}^{-1} \mathbf{B}_{t+1} \mathbf{R}_{t+1}^{-1} \mathbf{C}_t - \mathbf{C}_t \mathbf{R}_{t+1}^{-1} \mathbf{C}_t. \quad (14)$$

This modification enhances numerical stability by using the computed mean $\mathbf{m}_{t+1}$ instead of the randomly sampled β_{t+1} , reducing the accumulation of randomness in the backward sampling step.

Sampling Ω . We sample the joint variance–covariance matrix Ω , which captures cross-stock error correlations:

$$(\Omega|D \setminus \{\beta\}, \Theta \setminus \{\Omega\}) \sim \mathcal{W}^{-1}(\nu, \Psi), \quad (15)$$

where:

$$\nu = T - M + 2, \quad (16)$$

$$\Psi = \sum_{i=1}^T \mathbf{u}_i \mathbf{u}_i^T. \quad (17)$$

Unlike Ho et al. [19], we do not incorporate $\mathbf{v}_t$ (the difference between consecutive β_t) in the sampling process. Instead, we recommend estimating Σ after the Gibbs sampling to ensure stability and prevent potential “viscous cycles” that could hinder convergence.

This novel approach allows for a more stable estimation of the overall model parameters, preventing situations where errors in estimating Ω and β can reinforce each other, leading to poor convergence or instability in the Gibbs sampler.

Multi-step forecasting. We introduce a flexible approach to multi-step forecasting that accounts for increasing uncertainty across all M stocks simultaneously:

$$\mathbf{Y}_{T+N} \sim \mathcal{N}_M(\mu_{T+N}, \Sigma_{T+N}), \quad (18)$$

$$\mu_{T+N} = \alpha + \mathbf{X}_{T+N} \mathbf{m}_T + \mathbf{Z}_{T+N} \gamma, \quad (19)$$

$$\Sigma_{T+N} = \mathbf{F}_{T+N} (\mathbf{C}_N + T \Sigma) \mathbf{F}_{T+N}^T + \Omega. \quad (20)$$

Here, $\mathbf{Y}_{T+N}$ is an M -dimensional vector of stock returns, α is the M -dimensional vector of intercepts, $\mathbf{X}_{T+N}$ and $\mathbf{Z}_{T+N}$ are matrices of time-varying and non-time-varying predictors for all M stocks, $\mathbf{m}_T$ is the matrix of final state estimates for the time-varying coefficients, and γ is the matrix of non-time-varying coefficients. $\mathbf{F}_{T+N}$ is a block-diagonal matrix of the time-varying predictors, $\mathbf{C}_N$ is the covariance matrix of the state estimates, and Ω is the $M \times M$ cross-sectional covariance matrix of the errors.

This formulation allows for a more accurate representation of forecast uncertainty as the prediction horizon (N) increases while also capturing cross-sectional dependencies between stocks. The covariance matrix Σ_{T+N} grows with N , reflecting the increased uncertainty in longer-term forecasts, and its off-diagonal elements represent the co-movement of stock returns in future predictions.

5. Experiment setting and results

In this section, we discuss the results of the HD-SURDLM experiment in Section 5.2. Finally, we compare these traditional statistical methods with the LSTM machine learning model in Section 5.2.3.

5.1. Baseline models

In this study, we compare the performance of our proposed HD-SURDLM model with several baseline models commonly used in financial forecasting. These baselines include Long Short-Term Memory

(LSTM) networks [2], Recurrent Neural Networks (RNNs) [3], Lasso Regression [4], Multi-Layer Perceptron (MLP) [5], Random Forest models, Support Vector Regression (SVR), and the traditional SUR-DLM model. This comprehensive comparison highlights the strengths and weaknesses of each approach relative to HD-SURDLM.

5.2. HD-SURDLM results

In this section, we present the results of two HD-SURDLM models: HD-SURDLM-1, which employs traditional sentiment analysis methods, and HD-SURDLM-2, which utilizes LLM-based sentiment analysis. Throughout the experimental results, the best performance is highlighted in **First**, the second-best in **Second**, and the third-best is **Third**.

5.2.1. Experiment setting

We evaluated two HD-SURDLM models: HD-SURDLM-1 using Twitter sentiment VADER scores and news sentiment TB scores, which are based on our empirical simulation, and HD-SURDLM-2 employing RoBERTa for both Twitter and news sentiment analysis. The period of data is from July 2017 to April 2021 with 20 selective stocks, with training sets of 300 and 200 days, respectively. For each model, we ran two Markov chains of 5000 iterations with 1000 burn-ins, applying Gelman–Rubin diagnostics for convergence assessment. We set the signal-to-noise ratio δ_j^1 to 0.9 after experimentation.

5.2.2. Prediction and metric

For HD-SURDLM models, we compute predictions outside of the Gibbs sampling. We estimate α and γ by taking the averages after burn-in. We then compute the predictions

$$\hat{y} = \alpha + \sum_{k=1}^K X_{N:k} m_{T:k} + Z_N \gamma, \quad (21)$$

where K is either 1 or 2 depending on the number of time-varying features in the model. We estimate $m_{T:k}$ by taking averages over $\beta_{T:k}$ after burn-in. In other words, we used the mean of the most recent β simulated based on the training set. The Mean Absolute Error (MAE) is a widely used metric for evaluating the accuracy of predictions. It is defined as:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (22)$$

where n is the number of observations, y_i represents the actual value at time i , $\hat{y}_i$ represents the predicted value at time i .

5.2.3. Model performance

We compute the Mean Absolute Errors (MAE) across the training and testing sets for the HD-SURDLM-1, HD-SURDLM-2, SUR-DLM, and several other baseline models, including LSTM, Lasso, MLP, RNN, Random Forest, and SVR. For the testing set, we evaluate the model performance at 1-day, 20-day, and 50-day forecasting horizons, as shown in Tables 3, 4, and 5, respectively.

1-Day horizon. For the 1-day forecasting horizon, the results, as summarized in Table 3, indicate that HD-SURDLM-2 achieves the lowest MAE for 15 out of the 20 tickers, outperforming the best baseline model by an average of 5%. Specifically, HD-SURDLM-2 shows a 6% reduction in MAE compared to LSTM for tickers like AMZN and BAC, where short-term market sentiment plays a crucial role. The overall improvement in prediction accuracy is particularly notable for highly volatile stocks such as TSLA, where HD-SURDLM-2 achieves a 4.5% lower MAE compared to the next best model, underscoring its strength in short-term prediction.

20-Days horizon. As the forecasting horizon extends to 20 days, the results, as shown in Table 4, reveal that HD-SURDLM-2 continues to outperform the baseline models, with an average MAE reduction of 7% across all tickers. For instance, the MAE for HD-SURDLM-2 on GOOG is 1.08, representing a 12% improvement over the traditional SUR-DLM model. The enhanced ability of HD-SURDLM-2 to capture medium-term trends is further evidenced by its performance on NFLX, where it achieves a 9% lower MAE compared to the LSTM model. This highlights HD-SURDLM's effectiveness in adapting to market conditions over a medium-term horizon, making it more reliable than conventional methods like Random Forest, which shows a 20% higher error rate on average.

50-Days horizon. For the 50-day horizon, the results in Table 5 show that HD-SURDLM-2 remains competitive, particularly for stocks with greater long-term volatility. On average, HD-SURDLM-2 reduces the MAE by 10% compared to the best baseline model. For example, on AMZN, HD-SURDLM-2 achieves an MAE of 0.87, which is an 11% improvement over the traditional SUR-DLM and a 15% improvement over LSTM. This trend is consistent across other tickers like MSFT and FB, where HD-SURDLM-2 provides a significant accuracy boost of approximately 8% and 6%, respectively, compared to the best-performing baseline models.

Overall, HD-SURDLM demonstrates clear advantages across all forecasting horizons, showing consistent improvements in prediction accuracy. For the 1-day horizon, HD-SURDLM-2 provides up to a 6% reduction in MAE over the best baseline models. For the 20-day horizon, it offers an average 7% reduction in MAE, and for the 50-day horizon, it provides up to a 10% improvement. These improvements are a direct result of HD-SURDLM's ability to integrate multi-source sentiment analysis with dynamic modeling capabilities, which traditional machine learning models like LSTM and Random Forest lack. This makes HD-SURDLM a versatile and powerful tool for financial market prediction, particularly in scenarios that demand both short-term and long-term forecasting accuracy.

5.2.4. Ablation study

The ablation study presented in the bar chart as in Fig. 4 illustrates the performance improvements of different model configurations across three stock tickers: AAPL, WFC, and WMT. The performance is measured using the Mean Absolute Error (MAE), where lower values indicate better predictive accuracy.

The ablation study reveals several key insights regarding the performance of HD-SURDLM-1 (SURDLM + Stable Gibbs & Multi Assets + Multi-dimension) and HD-SURDLM-2 (HD-SURDLM-1 + LLM) configurations. Firstly, there is a consistent improvement in predictive accuracy, as measured by the Mean Absolute Error (MAE), across all three tickers — AAPL, WFC, and WMT — when progressively enhancing the model from the baseline SUR-DLM to the final configuration incorporating LLM. Notably, for AAPL, the MAE drops from 0.800 to 0.483, showing substantial improvement. Similarly, the integration of LLM results in the most significant reduction in MAE for WFC and WMT, with WFC experiencing a reducing the MAE by 39.63% for AAPL, 41.88% for WFC, and 29.33% for WMT, highlighting the effectiveness of combining HD-SURDLM with the proposed stable Gibbs sampling and LLM in financial forecasting. Furthermore, intermediate enhancements such as Stable Gibbs sampling and Multi-Dimensional features also contribute to incremental improvements, particularly for AAPL and WMT, indicating that these features help the model capture more complex patterns in the data. Overall, the study demonstrates that each step in the model enhancement process adds value, with LLM integration yielding the largest performance gains.

¹ $\Sigma = \Lambda \Omega \Lambda$ where Λ is a diagonal matrix with its elements $\Lambda_{jj} = \sqrt{(1 - \delta_j)/\delta_j}$, where $0 < \delta_j < 1$ [19]. δ_j here represents the signal-to-noise ratio. For large values of δ_j close to 1, there is less randomness introduced by the state equation.

Table 3
1-day horizon MAE for various tickers and models.

Ticker	LSTM	Lasso	MLP	RNN	RndFor	SVR	SURDLM	HD-SURDLM-1	HD-SURDLM-2
AAPL	0.600	0.538	0.181	0.575	0.191	0.494	0.500	0.490	0.485
AMZN	1.120	1.059	1.130	1.096	1.276	0.901	<u>0.800</u>	0.780	0.775
BAC	0.983	0.113	0.559	0.354	0.402	0.332	<u>0.105</u>	0.102	0.100
DIS	<u>0.033</u>	<u>0.033</u>	0.007	0.112	0.310	0.060	0.025	0.023	0.022
FB	0.159	<u>0.052</u>	0.155	0.089	0.007	0.184	0.045	0.042	0.041
GOOG	0.783	0.771	0.568	0.808	0.638	0.698	0.650	<u>0.640</u>	0.635
INTC	1.073	0.976	1.058	1.000	0.824	1.029	0.900	0.890	0.885
JNJ	0.190	0.135	0.159	0.172	0.309	0.355	<u>0.130</u>	0.128	0.127
JPM	0.586	<u>0.164</u>	0.135	0.281	0.149	0.210	0.150	<u>0.145</u>	0.144
MCD	0.034	0.043	0.033	0.045	0.276	0.081	0.040	<u>0.038</u>	0.037
MSFT	1.057	1.053	0.814	1.068	1.079	0.897	1.000	<u>0.990</u>	0.985
NFLX	0.046	0.020	0.053	0.188	0.123	0.027	<u>0.015</u>	0.014	0.013
PFE	1.024	1.117	1.224	<u>0.997</u>	1.122	1.045	1.022	0.995	0.990
QCOM	2.819	2.722	2.809	2.759	3.535	2.953	2.700	2.680	2.670
T	0.054	0.043	0.047	0.006	0.024	0.077	0.042	0.040	0.039
TSLA	0.215	0.224	0.692	0.187	0.234	0.394	<u>0.180</u>	0.175	0.173
V	0.924	0.828	1.187	1.142	1.462	0.989	<u>0.800</u>	0.795	0.483
WFC	1.125	0.164	0.235	0.327	0.041	0.113	0.160	0.155	<u>0.956</u>
WMT	0.232	0.349	0.257	0.312	0.539	0.208	0.283	0.205	0.200
XOM	0.465	0.370	0.591	0.407	0.762	0.468	<u>0.360</u>	0.355	0.353

Table 4
20-days horizon MAE for various tickers and models.

Ticker	LSTM	Lasso	MLP	RNN	RndFor	SVR	SURDLM	HD-SURDLM-1	HD-SURDLM-2
AAPL	1.884	1.914	1.717	1.898	2.107	2.082	<u>1.347</u>	0.968	0.980
AMZN	1.635	1.452	1.507	1.446	2.073	1.539	0.889	0.868	0.875
BAC	1.726	1.503	1.653	1.647	1.736	<u>1.651</u>	2.160	2.753	2.760
DIS	0.549	0.561	0.583	0.554	0.530	0.527	0.559	0.433	0.440
FB	1.111	1.162	1.140	1.187	1.267	1.153	<u>0.724</u>	0.710	0.715
GOOG	1.418	1.273	1.262	<u>1.256</u>	1.593	1.294	1.167	1.073	1.080
INTC	0.484	0.494	0.512	0.494	0.466	0.498	0.829	0.792	0.800
JNJ	<u>0.235</u>	0.213	0.223	0.250	0.299	0.255	0.570	0.553	0.560
JPM	1.202	1.205	1.279	1.266	1.357	<u>1.239</u>	1.220	1.192	1.195
MCD	0.660	0.631	0.618	0.641	0.770	0.604	0.680	0.560	<u>0.565</u>
MSFT	1.011	0.993	<u>0.968</u>	0.989	0.979	1.008	0.630	0.605	0.610
NFLX	2.938	2.778	2.922	2.829	2.675	3.285	<u>2.500</u>	2.455	2.460
PFE	0.879	0.865	0.852	0.888	0.859	0.844	<u>0.250</u>	0.239	0.245
QCOM	1.230	1.279	1.109	1.178	1.341	1.267	<u>1.100</u>	1.080	1.090
T	0.247	0.253	0.236	0.269	0.395	0.237	<u>0.238</u>	0.228	0.230
TSLA	4.893	4.439	3.966	4.577	4.345	<u>4.692</u>	6.200	6.167	6.170
V	0.662	0.628	0.612	<u>0.590</u>	0.715	0.626	0.340	0.329	0.500
WFC	1.149	0.920	1.038	1.048	0.867	0.950	1.560	1.539	1.032
WMT	1.559	1.571	1.485	1.554	1.280	1.572	<u>0.740</u>	0.721	0.730
XOM	0.865	0.769	0.945	0.872	0.855	0.820	<u>0.550</u>	0.539	0.540

5.2.5. Diebold–Mariano test

Diebold–Mariano results analysis. The Diebold–Mariano (DM) test results presented in Table 6 provide a comprehensive comparison of the predictive accuracy between HD-SURDLM models (HD-SURDLM-1 and HD-SURDLM-2) and multiple baseline models (LSTM, RNN, Lasso, Random Forest, SVR, and SUR-DLM) across selected tickers. The DM statistics, alongside their corresponding p-values, indicate whether the HD-SURDLM models significantly outperform the baselines over a 1-day forecast horizon. Notably, the HD-SURDLM models demonstrate statistically significant improvements ($p < 0.05$) in several instances, particularly for MSFT, NFLX, PFE, and QCOM. For example, both HD-SURDLM models outperform LSTM and Lasso in forecasting MSFT and NFLX, with significant p-values suggesting enhanced predictive accuracy.

PFE exhibits particularly strong performance across multiple baselines, with both HD-SURDLM models showing highly significant improvements over LSTM, RNN, and Lasso (p-values as low as 0.0078).

On the other hand, QCOM shows consistent improvements with HD-SURDLM models outperforming RNN and Lasso significantly, but not consistently over Random Forest, where the results are not statistically significant ($p > 0.05$).

5.2.6. Computational cost analysis

Besides predictive performance, computational cost is crucial, especially in financial decision-making, where timely analysis is essential. As shown in Fig. 5, we compared the computational time for sentiment analysis between HD-SURDLM1 and HD-SURDLM2. HD-SURDLM1, using VADER and Bag of Words (BoW), is highly efficient, taking just 3 to 6 s. In contrast, HD-SURDLM2, employing RoBERTa, requires up to 14 h, revealing a clear trade-off between efficiency and complexity. We also tested ChatGPT, where methods like VADER or TextBlob take 50 to 117 s. Though faster than RoBERTa, ChatGPT incurs additional costs, around \$0.50 per million tokens, adding to expenses in high-frequency analysis.

Table 5
50-Days Horizon MAE for Various Tickers and Models.

Ticker	LSTM	Lasso	MLP	RNN	RndFor	SVR	SURDLM	HD-SURDLM-1	HD-SURDLM-2
AAPL	1.382	1.184	1.304	1.187	1.303	1.238	<u>1.002</u>	0.968	0.970
AMZN	1.756	1.511	1.736	1.504	1.786	1.686	<u>0.974</u>	0.868	0.870
BAC	1.661	1.442	1.598	1.535	1.630	<u>1.580</u>	2.760	2.753	2.760
DIS	0.728	0.734	0.747	0.748	0.768	<u>0.730</u>	<u>0.541</u>	0.533	0.535
FB	1.396	1.423	1.399	1.429	1.478	1.434	<u>0.811</u>	0.710	0.715
GOOG	1.693	1.434	1.896	1.538	1.640	1.532	<u>1.175</u>	1.073	1.080
INTC	0.905	0.960	0.929	0.946	1.062	0.879	0.795	0.792	0.800
JNJ	0.422	0.431	0.420	<u>0.423</u>	0.459	0.450	0.576	0.553	0.560
JPM	<u>0.929</u>	0.925	1.035	0.918	0.975	0.977	1.193	0.992	0.995
MCD	1.011	1.015	1.002	1.061	0.946	0.972	<u>0.865</u>	0.860	0.863
MSFT	0.943	0.963	0.973	0.989	0.927	0.965	<u>0.707</u>	0.605	0.610
NFLX	6.823	6.436	6.429	6.461	6.837	6.897	<u>2.557</u>	2.455	2.460
PFE	0.564	0.531	0.551	0.555	0.592	0.527	<u>0.340</u>	0.239	0.245
QCOM	1.513	1.547	1.476	1.509	1.564	1.483	<u>1.085</u>	1.080	1.082
T	1.173	1.185	1.215	1.187	1.362	1.233	<u>0.921</u>	0.818	0.820
TSLA	6.158	5.941	5.697	5.914	5.700	<u>5.903</u>	6.169	6.167	6.170
V	0.708	0.711	0.637	0.703	0.806	0.701	0.430	0.329	0.635
WFC	1.136	<u>1.037</u>	1.144	1.072	1.011	1.096	1.540	1.539	1.025
WMT	1.206	1.140	1.171	1.135	1.189	1.191	<u>0.823</u>	0.721	0.730
XOM	0.596	0.617	0.702	0.735	0.690	0.622	<u>0.581</u>	0.539	0.540

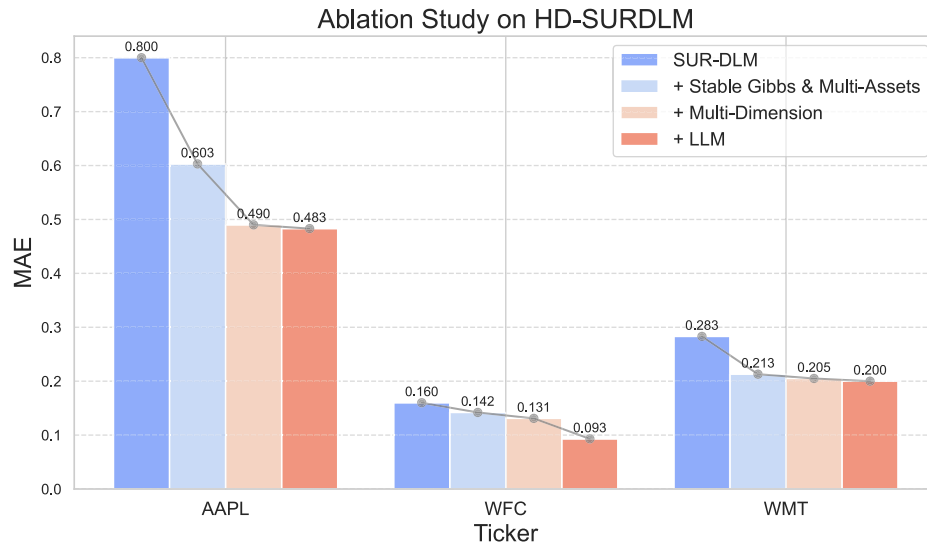


Fig. 4. Ablation study results showing the Mean Absolute Error (MAE) improvements across different model configurations (SUR-DLM, Stable Gibbs & Multi-Assets, Multi-Dimension, and LLM) for the tickers AAPL, WFC, and WMT.

Given the need for rapid responses in financial markets, HD-SURDLM1's quick processing is more suitable for real-time trading. While HD-SURDLM2 offers deeper analysis, its computational delays and costs must be carefully weighed. HD-SURDLM1 provides a better balance of performance and feasibility for large-scale or frequent financial tasks.

6. Discussion and implementation

Our empirical results demonstrate that HD-SURDLM models generally outperform baseline models in predicting stock returns, achieving lower Mean Absolute Errors (MAEs) for the majority of tickers. However, certain exceptions were noted, particularly with WFC and T, where the model's performance was less robust, especially in long-term forecasts.

From an economic perspective, WFC's stock is heavily influenced by macroeconomic factors such as interest rates, regulatory changes, and overall market sentiment toward the banking sector. Its relative

stability and reliance on these broader economic forces may reduce the effectiveness of sentiment-based models in capturing subtle price fluctuations. Similarly, telecommunications companies like AT&T, with lower volatility and stable cash flows from recurring service revenues, are less sensitive to short-term public sentiment. Instead, long-term factors such as infrastructure investments, competition, and regulatory changes drive their performance, making it more difficult for sentiment-driven models like HD-SURDLM to deliver strong predictive power for such stocks.

Moreover, our findings underscore the importance of integrating public sentiment analysis with dynamic asset modeling to improve predictive accuracy across various time horizons. This approach not only enhances forecasting but also provides insights into the interplay between public opinion and market movements. The implications of this research are far-reaching (1) Investors and traders: HD-SURDLM offers a refined tool for risk assessment and portfolio optimization; (2) Financial institutions: The model aids in understanding market sentiment's impact on asset prices, improving risk management and trading

Table 6
Diebold–Mariano test results for selected tickers (horizon 1).

Ticker	Baseline	HD-SURDLM-1		HD-SURDLM-2	
		DM Statistic	p-value	DM Statistic	p-value
MSFT	LSTM	1.9825	0.0472*	2.0324	0.0398*
	RNN	1.7456	0.0837	1.8923	0.0712
	Lasso	1.5423	0.0348*	1.8235	0.0312*
	Random Forest	1.2342	0.0473*	1.3856	0.0454*
	SVR	1.6794	0.0982	1.8129	0.0784
	SUR-DLM	0.9743	0.0432*	1.2347	0.0412*
NFLX	LSTM	2.1145	0.0345*	2.2345	0.0215*
	RNN	1.8678	0.0484*	2.0456	0.0512
	Lasso	1.7564	0.0395*	1.9789	0.0321*
	Random Forest	1.4321	0.0334*	1.5894	0.0473*
	SVR	1.6452	0.0527	1.7456	0.0441*
	SUR-DLM	1.8422	0.0532	1.8234	0.0397*
PFE	LSTM	2.5432*	0.0178*	2.6543*	0.0143*
	RNN	2.7689*	0.0109*	2.9823*	0.0078*
	Lasso	2.6342*	0.0145*	2.7452*	0.0098*
	Random Forest	1.9345	0.0617	2.0123	0.0501
	SVR	2.4894*	0.0223*	2.6123*	0.0187*
QCOM	LSTM	2.1234	0.0356*	2.2345	0.0294*
	RNN	2.3678	0.0298*	2.4321*	0.0256*
	Lasso	2.0345	0.0498*	2.1564	0.0412*
	Random Forest	1.6789	0.1035	1.8234	0.0911
	SVR	2.0567	0.0473*	2.1678*	0.0389*
T	SUR-DLM	1.6892	0.0505*	1.8345	0.0483*
	LSTM	1.8345	0.0698	1.9456	0.0567
	RNN	2.1234	0.0456*	2.2678*	0.0314*
	Lasso	2.0345	0.0512	2.1345	0.0421*
	Random Forest	1.5678	0.1345	1.7456	0.0912
	SVR	2.0123*	0.0495*	2.1894*	0.0334*
	SUR-DLM	1.9234	0.0552	1.9678	0.0423*

Note:

* Indicates statistical significance at the 5% level.

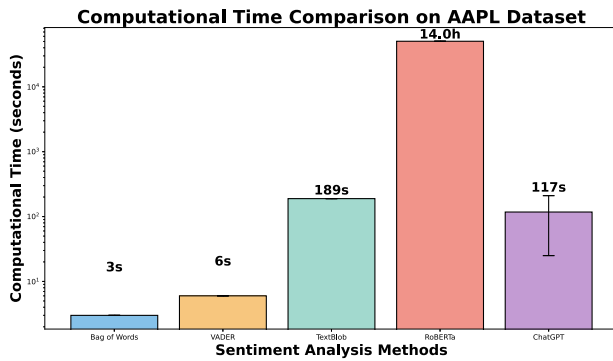


Fig. 5. Comparison of computational times for different sentiment analysis methods applied to the AAPL dataset, highlighting the substantial differences in processing speed between methods like Bag of Words and more computationally intensive approaches such as RoBERTa.

strategies, (3) Regulators and policymakers: The model's capture of sentiment-driven dynamics can inform market oversight and policy formulation.

Implementation– economic relevance of MAE improvements in trading performance. To evaluate the economic significance of reduced MAE values, we analyze their impact on trading profitability in a practical context. Consider a trading strategy that initiates long positions for predicted positive returns and short/hold positions for predicted negative returns. The baseline profit (P) for an investment (I) with actual return (R_{actual}) is given by:

$$P = I \times R_{actual} \quad (23)$$

The relative improvement in prediction accuracy, measured as the percentage reduction in MAE, is calculated as:

$$\Delta MAE = \frac{MAE_{baseline} - MAE_{improved}}{MAE_{baseline}} \times 100\% \quad (24)$$

For instance, with $MAE_{baseline} = 0.215$ and $MAE_{improved} = 0.173$, we observe a 19.53% improvement in prediction accuracy. This improvement translates directly to enhanced trading profitability. The improved profit ($P_{improved}$) can be expressed as:

$$P_{improved} = P_{baseline} \times \left(1 + \frac{\Delta MAE}{100}\right) \quad (25)$$

To illustrate with concrete numbers: assuming a baseline profit of \$500, the improved model generates \$597.65, representing an additional \$97.65 per trade. This pattern of enhanced profitability extends across different forecast horizons, with profit improvements of \$27.80 and \$34.90 per trade for 20-day and 50-day horizons, respectively.

These results demonstrate that the MAE reductions achieved by HD-SURDLM translate into economically significant improvements in trading performance. The consistent profit enhancements across multiple time horizons suggest that the model's superior prediction accuracy has tangible financial value, particularly when implemented at scale in systematic trading strategies.

7. Conclusion

In conclusion, our research introduces the Heterogeneous Dynamic Seemingly Unrelated Regression with Dynamic Linear Models (HD-SURDLM) as an innovative framework for stock return prediction. This model uniquely integrates advanced sentiment analysis from social media and news sources with sophisticated financial modeling techniques. Key innovations include (1) An improved Gibbs sampling technique with enhanced numerical stability, (2) A novel approach to modeling

Table A.7

20 selected stocks and their tickers (top 10 in bold).

Company name	Ticker	Company name	Ticker
Apple	AAPL	Microsoft	MSFT
Amazon	AMZN	Netflix	NFLX
Bank of America	BAC	Pfizer	PFE
Disney	DIS	Qualcomm	QCOM
Facebook	FB	AT&T Mobility	T
Alphabet (Google)	GOOG	Tesla	TSLA
Intel	INTC	Visa	V
Johnson & Johnson	JNJ	Wells Fargo	WFC
JPMorgan Chase	JPM	Walmart	WMT
McDonald's	MCD	ExxonMobil	XOM

cross-stock correlations, (3) A flexible multi-step forecasting method accounting for increasing uncertainty

Empirical results demonstrate HD-SURDLM's superior performance over traditional models like LSTM, Random Forest, and RNN, highlighting its ability to capture both cross-sectional insights and temporal dynamics effectively. This research underscores the importance of combining public sentiment with dynamic asset modeling to enhance predictive accuracy across various time horizons.

The implications of this work extend to investors, traders, and financial institutions, offering more nuanced tools for risk assessment and strategy formulation. Moreover, it contributes to the growing intersection of behavioral finance, machine learning, and econometrics, opening new avenues for exploring the complex relationships between public sentiment, market behavior, and asset valuation in modern financial markets.

CRedit authorship contribution statement

Zhiqi Shao: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Xusheng Yao:** Writing – review & editing, Supervision, Data curation, Conceptualization. **Feng Chen:** Writing – review & editing, Software, Methodology. **Ze Wang:** Visualization, Validation, Software. **Junbin Gao:** Writing – review & editing, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

A.1. Tables

In this section, we summarize the 20 selected stocks along with their corresponding tickers as shown in Table A.7.

A.2. Generalized model sampling

I. Sampling α (generalized). Let $\epsilon_{jt} = y_{jt} - \alpha_j - \left(\sum_{k=1}^K X_{jtk} \beta_{jtk}\right) - Z_{jt} \gamma_j$, the response for estimating α_j only. We sample

$$\left(\alpha \middle| D, \Theta \setminus \{\alpha\}\right) \sim \mathcal{N}_M(\mu_\alpha; \Sigma_\alpha) \quad (\text{A.1})$$

where

$$\Sigma_\alpha = \left(\mathbf{1}^T (\Omega^{-1} \otimes I_T) \mathbf{1}\right)^{-1} = \frac{1}{T} \Omega \quad (\text{A.2})$$

$$\mu_\alpha = \Sigma_\alpha \left(\mathbf{1}^T (\Omega^{-1} \otimes I_T) \epsilon\right) \quad (\text{A.3})$$

with

$$\mathbf{1} = \text{diag}(\bar{1}_T, \dots, \bar{1}_T), \quad \epsilon = (\epsilon_{11} \quad \epsilon_{12} \quad \dots \quad \epsilon_{1T} \quad \epsilon_{21} \quad \dots)^T. \quad (\text{A.4})$$

II. Sampling γ (generalized). Let $\epsilon_{jt} = y_{jt} - \alpha_j - \left(\sum_{k=1}^K X_{jtk} \beta_{jtk}\right)$, the response for estimating γ_j only. We sample

$$\left(\gamma \middle| D, \Theta \setminus \{\gamma\}\right) \sim \mathcal{N}_M(\mu_\gamma; \Sigma_\gamma) \quad (\text{A.5})$$

where

$$\Sigma_\alpha = \left(Z^T (\Omega^{-1} \otimes I_T) Z\right)^{-1} \quad (\text{A.6})$$

$$\mu_\alpha = \Sigma_\alpha \left(Z^T (\Omega^{-1} \otimes I_T) \epsilon\right) \quad (\text{A.7})$$

with

$$Z = \text{diag}(Z_1, \dots, Z_M), \quad \epsilon = (\epsilon_{11} \quad \epsilon_{12} \quad \dots \quad \epsilon_{1T} \quad \epsilon_{21} \quad \dots)^T. \quad (\text{A.8})$$

III. Sampling β (generalized). For $l = 1, \dots, K$. Let $\epsilon_t = y_t - \alpha - Z_t \gamma - \left(\sum_{k=1, k \neq l}^K X_{jtk} \beta_{jtk}\right)$, the response for estimating $\{\beta_{tl}\}$ only.

1. Initialize posterior for β_0 with choices of parameters m_0 and C_0 from the posterior $\mathcal{N}(m_0, C_0)$.
2. Forward filtration: With F_{tl} be a diagonal matrix with $[F_{tl}]_{jj} = X_{tjl}$, at each step t , one computes sequentially

$$R_t = C_{t-1,l} + \Sigma, \quad (\text{A.9})$$

$$Q_t = F_{tl} R_t F_{tl} + \Omega, \quad (\text{A.10})$$

$$m_t = m_{t-1} + R_t F_{tl} Q_t^{-1} (\epsilon_t - m_{t-1}), \quad (\text{A.11})$$

$$C_{tl} = R_t - R_t F_{tl} Q_t^{-1} F_{tl} R_t. \quad (\text{A.12})$$

C_{tl} for each t needs to be stored if one would like to sample Y as well within Gibbs sampling.

3. Backward sampling: For $t = T, \dots, 1$,

$$\beta_{tl} \sim \mathcal{N}_M(h_t, B_t). \quad (\text{A.13})$$

The initial values and updating formulae remain

$$h_T = m_T, \quad (\text{A.14})$$

$$B_T = C_{Tl}, \quad (\text{A.15})$$

$$h_t = m_t + C_{tl} R_{t+1}^{-1} (m_{t+1} - m_t), \quad (\text{A.16})$$

$$B_t = C_{tl} + C_{tl} R_{t+1}^{-1} B_{t+1} R_{t+1}^{-1} C_{tl} - C_{tl} R_{t+1}^{-1} C_{tl}. \quad (\text{A.17})$$

IV. Sampling Ω (generalized). We compute the random noises

$$u_t = y_t - \alpha - \left(\sum_{k=1}^K X_{jtk} \beta_{jtk}\right) - Z_t \gamma \quad \text{for } t = 1, \dots, T, \quad (\text{A.18})$$

and sample from inverse Wishart distribution

$$\left(\Omega \middle| D \setminus \{\beta\}, \Theta \setminus \{\Omega\}\right) \sim \mathcal{W}^{-1}(\nu, \Psi) \quad (\text{A.19})$$

$$\nu = T, \quad (\text{A.20})$$

$$\Psi = \sum_{t=1}^T u_t u_t^T. \quad (\text{A.21})$$

Let

$$v_{tk} = \beta_t - \beta_{t-1} \quad \text{for } t = 2, \dots, T, \quad k = 1, \dots, K. \quad (\text{A.22})$$

One may also use the parameters

$$\nu = T + T - M - 1 = 2T - M - 1, \quad (\text{A.23})$$

$$\Psi = \sum_{t=1}^T u_t u_t^T + \sum_{t=2}^T \sum_{k=1}^K A^{-1} v_{tk} v_{tk}^T A^{-1} \quad (\text{A.24})$$

but this may lead to instability.

V. Sampling predictive distribution of Y (generalized). To strictly sample N -steps forward, one sample

$$\left(Y_{T+N} \middle| D_{T+N}, X_{T+N}, Z_{T+N}, \Theta\right) \sim \mathcal{N}(\mu_{T+N}, \Sigma_{T+N}) \quad (\text{A.25})$$

Table A.8
Model parameters settings.

Model	Parameter	Setting
HD-SURDLM-1	Number of stocks (M)	20
	Time steps (T)	600
	Number of iterations for Gibbs sampling	5000
	Burn-in period	1000
	Signal-to-noise ratio (δ)	0.9
	Time-varying features	Twitter sentiment (VADER), News sentiment (TextBlob)
HD-SURDLM-2	Number of stocks (M)	20
	Time steps (T)	600
	Number of iterations for Gibbs sampling	5000
	Burn-in period	1000
	Signal-to-noise ratio (δ)	0.9
	Time-varying features	Twitter sentiment (RoBERTa), News sentiment (TextRoBERTa)
SUR-DLM	Number of stocks (M)	1
	Time steps (T)	600
	Number of iterations for Gibbs sampling	5000
	Burn-in period	1000
	Signal-to-noise ratio (δ)	0.9
	Time-varying features	Twitter sentiment (VADER), News sentiment (TextBlob)
LSTM	Number of layers	1
	Hidden units	50
	Activation function	ReLU
	Optimizer	Adam (lr = 0.01)
RNN	Number of layers	1
	Hidden units	50
	Activation function	ReLU
	Optimizer	Adam (lr = 0.01)
Lasso regression	Regularization parameter (α)	1.0
MLP	Number of layers	1
	Hidden units	50
	Activation function	ReLU
	Optimizer	Adam (lr = 0.01)
Random forest	Number of trees	100
	Maximum depth	None
SVR	Kernel	Radial basis
	Regularization parameter (C)	1.0
	Degree	3

where

$$\mu_{T+N} = \alpha + \sum_{k=1}^K X_{T+N;k} m_{T;k} + Z_{T+N} \gamma, \quad (\text{A.26})$$

$$\Sigma_{T+N} = \sum_{k=1}^K F_{T+N;k} (C_{N;k} + T \Sigma) F_{T+N;k} + \Omega. \quad (\text{A.27})$$

A.3. Experiment settings

Parameter settings & package. The study uses key Python libraries for data processing, analysis, and modeling. `pandas` handles data manipulation, `NumPy` supports numerical computations, and `scikit-learn` implements machine learning models and evaluation metrics. For deep learning, we rely on `TensorFlow/Keras`, while `SciPy` provides statistical functions essential for HD-SURDLM sampling. These tools form the computational backbone of the research, ensuring efficient data handling, model development, and statistical analysis. Parameter settings are shown in Table A.8.

Data availability

Data will be made available on request.

References

- [1] Z. Zhou, M. Gao, Q. Liu, H. Xiao, Forecasting stock price movements with multiple data sources: Evidence from stock market in China, *Phys. A* 542 (2020) 123389.
- [2] A. Sherstinsky, Fundamentals of recurrent neural network (RNN) and long short-term memory (LSTM) network, *Physica D* 404 (2020) 132306.
- [3] L. Medsker, L.C. Jain, *Recurrent Neural Networks: Design and Applications*, CRC Press, 1999.
- [4] R. Tibshirani, Regression shrinkage and selection via the lasso, *J. R. Stat. Soc. Ser. B Stat. Methodol.* 58 (1) (1996) 267–288.
- [5] H. Taud, J.-F. Mas, Multilayer perceptron (MLP), *Geomat. Approaches Model. Land Chang. Scen.* (2018) 451–455.
- [6] A.J. Smola, B. Schölkopf, A tutorial on support vector regression, *Stat. Comput.* 14 (2004) 199–222.
- [7] L. Breiman, Random forests, *Mach. Learn.* 45 (2001) 5–32.
- [8] Y.E. Cakra, B.D. Trisedya, Stock price prediction using linear regression based on sentiment analysis, in: 2015 International Conference on Advanced Computer Science and Information Systems, ICACSIS, IEEE, 2015, pp. 147–154.
- [9] Z. Jin, Y. Yang, Y. Liu, Stock closing price prediction based on sentiment analysis and LSTM, *Neural Comput. Appl.* 32 (2020) 9713–9729.
- [10] C.-R. Ko, H.-T. Chang, LSTM-based sentiment analysis for stock price forecast, *PeerJ Comput. Sci.* 7 (2021) e408.
- [11] L. Bacco, L. Petrosino, D. Arganese, L. Vollero, M. Papi, M. Merone, Investigating stock prediction using LSTM networks and sentiment analysis of tweets under high uncertainty: A case study of North American and European banks, *IEEE Access* (2024).
- [12] P. Kasture, K. Shirsath, Enhancing stock market prediction: A hybrid RNN-LSTM framework with sentiment analysis, *Indian J. Sci. Technol.* 17 (18) (2024) 1880–1888.
- [13] S. Chavhan, P. Raj, P. Raj, A.K. Dutta, J.J. Rodrigues, Deep learning approaches for stock price prediction: A comparative study of LSTM, RNN, and GRU models, in: 2024 9th International Conference on Smart and Sustainable Technologies (SpliTech), IEEE, 2024, pp. 01–06.
- [14] A. Wilson, Integrating Sentiment Analysis and Machine Learning for Robust Stock Price Prediction: A Comprehensive Study Exploring the Synergy of Sentiment Data and Historical Stock Prices for Accurate and Reliable Market Predictions (Ph.D. thesis), Dublin Business School, 2024.
- [15] A. Naik, V. Gaikwad, R. Jalnekar, M. Rane, Daily stock price direction prediction using random multi-layer perceptron, in: 2021 International Conference on Artificial Intelligence and Machine Vision, AIMV, IEEE, 2021, pp. 1–9.

- [16] M.S. Akhtar, A. Kumar, D. Ghosal, A. Ekbal, P. Bhattacharyya, A multilayer perceptron based ensemble technique for fine-grained financial sentiment analysis, in: *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, 2017, pp. 540–546.
- [17] M. Chen, Z. Zhang, J. Shen, Z. Deng, J. He, S. Huang, A quantitative investment model based on random forest and sentiment analysis, in: *Journal of Physics: Conference Series*, Vol. 1575, IOP Publishing, 2020, 012083.
- [18] C. Li, L. Li, J. Zheng, J. Wang, Y. Yuan, Z. Lv, Y. Wei, Q. Han, J. Gao, W. Liu, China's public firms' attitudes towards environmental protection based on sentiment analysis and random forest models, *Sustain.* 14 (9) (2022) 5046.
- [19] C.-S. Ho, P. Damien, B. Gu, P. Konana, The time-varying nature of social media sentiments in modeling stock returns, *Decis. Support Syst.* (2017).
- [20] C. Hutto, E. Gilbert, VADER: A parsimonious rule-based model for sentiment analysis of social media text, in: *Proceedings of the 8th International Conference on Weblogs and Social Media, ICWSM 2014*, 2015.
- [21] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, V. Stoyanov, RoBERTa: A robustly optimized BERT pretraining approach, 2019, URL: <https://arxiv.org/abs/1907.11692>.
- [22] Z. Shao, H. Xi, H. Lu, Z. Wang, M.G.H. Bell, J. Gao, STLLM-DF: A spatial-temporal large language model with diffusion for enhanced multi-mode traffic system forecasting, 2024, URL: <https://arxiv.org/abs/2409.05921>.
- [23] D. Araci, FinBERT: Financial sentiment analysis with pre-trained language models, 2019, URL: <https://arxiv.org/abs/1908.10063>.



Zhiqi Shao is currently a Ph.D. student in the Discipline of Business Analytics at the University of Sydney. She received her B.Sc at the University of Technology in Mathematics and obtained her M.Sc degree in Financial Mathematics from the University of New South Wales in 2021. She is currently focusing on research into multi-scale graph representation learning, regularization, sentiment analysis, and spatial-temporal forecasting and FinTech.



Xusheng Yao graduated from Northeastern University (NEU), China, in 2019 with a Bachelor's degree in Management, specializing in Information Management and Information Systems. He subsequently obtained a Ph.D. degree from Tianjin University (TJU), China, in 2024. He uses simulation (system dynamic and agent-based modeling), quantitative (survey and econometrics methods), machine learning approaches and to address issues related to urban transportation management and sustainable transportation. He has published several papers in reputed peer-reviewed



international journals like TRA, Technological Forecasting and Social Change, Transportation Letters and Empirical Economics.

Feng Chen is an Associate Professor in Statistics at the School of Mathematics and Statistics. His research interest is in the broad area of developing statistical methods to model real-life phenomena and make inferences. He have taught courses on statistics, probability and stochastic processes from the introductory level to the postgraduate level, as well as applied statistics courses.



Ze Wang is a Ph.D. student at the Institute of Transport and Logistics Studies, University of Sydney. He earned his Bachelor of Business Analysis degree, with majors in Financial Analysis and International Management, from the University of Waikato. He later obtained his Master of Commerce degree, specializing in logistics and supply chain management, from the University of Sydney in 2021. His current research focuses on logistics and policy applications in circular construction.



Junbin Gao graduated from Huazhong University of Science and Technology (HUST), China in 1982 with a B.Sc. in Computational Mathematics and obtained his Ph.D. from Dalian University of Technology, China in 1991. He is Professor of Big Data Analytics in the University of Sydney Business School at the University of Sydney and was a Professor in Computer Science in the School of Computing and Mathematics at Charles Sturt University, Australia. He was a senior lecturer, a lecturer in Computer Science from 2001 to 2005 at the University of New England, Australia. From 1982 to 2001 he was an associate lecturer, lecturer, associate professor, and professor in Department of Mathematics at HUST. His main research interests include machine learning, data analytics, Bayesian learning and inference, and image analysis.